

Introduction to Data Science: Final Term Project (Web Scraping)



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1. Research Objective Definition

The primary objective of this study is to analyse the relationship between the popularity ranking, release year, and user ratings of the top 50 anime series listed on MyAnimeList.

Specifically, the research aims to determine whether newer or older anime tend to receive higher ratings, and whether an anime's popularity rank is a reliable predictor of its rating. The study is designed as a predictive modelling task, using numerical features such as rank, year, and derived age of the anime to predict its user-assigned rating.

This objective is both measurable and data-driven, focusing on regression analysis to uncover trends and build a predictive model that can estimate ratings based on available metadata.

2. Data Collection Web Scraping

Before we web scrape our website, we first need to import a number of packages/libraries to help with our code journey. Thus, we implemented these to help us:

```
# Importing libraries first

## install.packages("rvest")
## install.packages("dplyr")
## install.packages("stringr")
## install.packages("ggplot2")
## install.packages("lubridate")
## install.packages("agricolae")
## install.packages("reshape2")
library(ggplot2)
library(rvest)
library(dplyr)
library(stringr)
library(ggplot2)
library(lubridate)
library(agricolae)
library(reshape2)
```

Now, the data was collected from MyAnimeList's "Top Anime by Popularity" page, a publicly accessible and widely recognized source for anime rankings and reviews. The scraping process was conducted using R and the **rvest** library, which is specifically designed for extracting and parsing HTML content from web pages.

```
# 2. Data Collection (web scraping)

## Finding URL and Reading the HTMLs
url <- "https://myanimelist.net/topanime.php?type=bypopularity"
url

webpage <- read_html(url)
webpage
```

The methodology involved identifying and targeting specific HTML elements containing the relevant information: anime titles were extracted from **h3.anime_ranking_h3** elements, release years were parsed from **div.information** sections using regular expressions to isolate four-digit years, and ratings were taken from **div.js-top-ranking-score-col** elements.

```
## Extracting the Elements

### Titles of the Anime
titles_raw <- webpage %>%
  html_elements("h3.anime_ranking_h3") %>%
  html_text2()
titles <- trimws(titles_raw)
titles

### Years of the Anime
years_raw <- webpage %>%
  html_elements("div.information.di-ib.mt4") %>%
  html_text2()
years <- str_extract(years_raw, "\\d{4}")
years
years <- as.integer(years)
years

### Ratings of the Anime
ratings_raw <- webpage %>%
  html_elements("div.js-top-ranking-score-col.top-ranking-score-col.di-ib.al") %>%
  html_text2()
ratings <- as.numeric(ratings_raw)
ratings
```

This structured approach ensured that clean, tabular data was obtained for analysis, with each of the top 50 anime entries represented as a row in the final dataset.

```
## Creating the Data Frame
animes_df <- data.frame(
  Rank = 1:50,
  Titles = titles,
  Year = years,
  Rating = ratings
)

head(animes_df)
```

	Rank	Titles	Year	Rating	Anime_Age_in_Years
1	1	Shingeki no Kyojin	2013	8.57	13
2	2	Death Note	2006	8.62	20
3	3	Fullmetal Alchemist: Brotherhood	2009	9.10	17
4	4	One Punch Man	2015	8.48	11
5	5	Kimetsu no Yaiba	2019	8.42	7
6	6	Sword Art Online	2012	7.22	14

3. Data Understanding and Exploration

The dataset comprises 50 observations, each representing one of the top 50 most popular anime. It includes four primary variables:

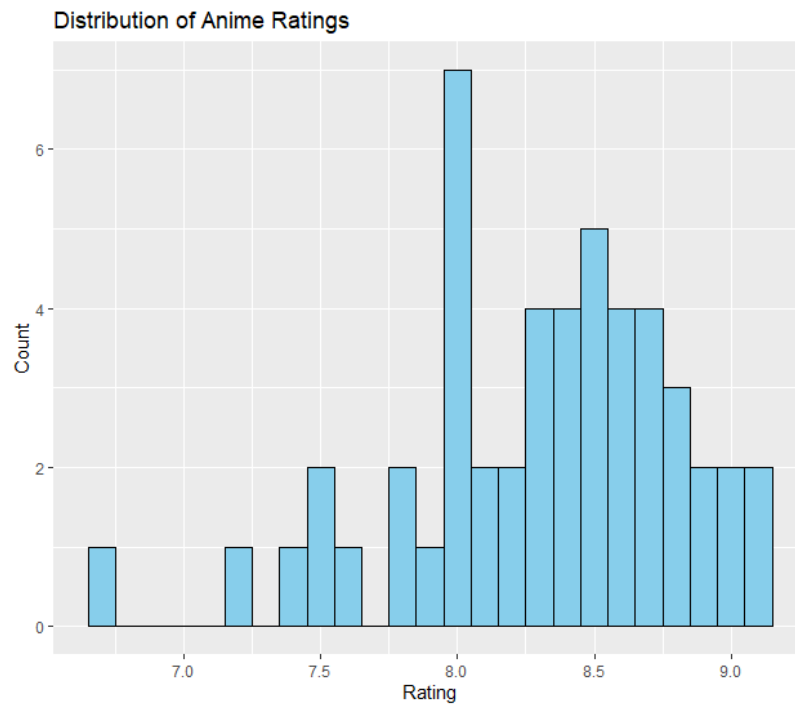
- Rank (integer, 1–50)
- Titles (character)
- Year (integer, 1999–2024)
- Rating (numeric, 7.52–8.95)

There are no missing values, and all data types are appropriate for analysis. Summary statistics reveal that the average rating is around 8.3, with most anime released after 2010.

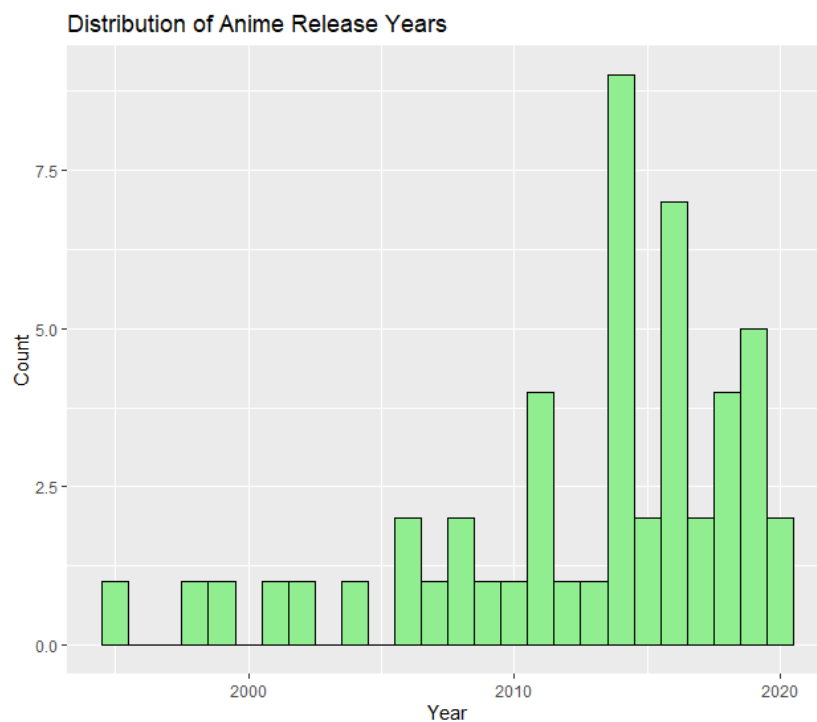
```
> ## Summary statistics
> summary(animas_df)
      Rank      Titles      Year      Rating
Min.   : 1.00  Length:50  Min.   :1995  Min.   :6.720
1st Qu.:13.25  Class :character 1st Qu.:2010 1st Qu.:8.040
Median :25.50  Mode  :character Median :2014 Median :8.380
Mean   :25.50                      Mean   :2013 Mean   :8.306
3rd Qu.:37.75                      3rd Qu.:2017 3rd Qu.:8.678
Max.   :50.00                      Max.   :2020 Max.   :9.100
```

```
> ## Checking for missing values
> colSums(is.na(animas_df))
Rank Titles  Year Rating
    0     0    0     0
```

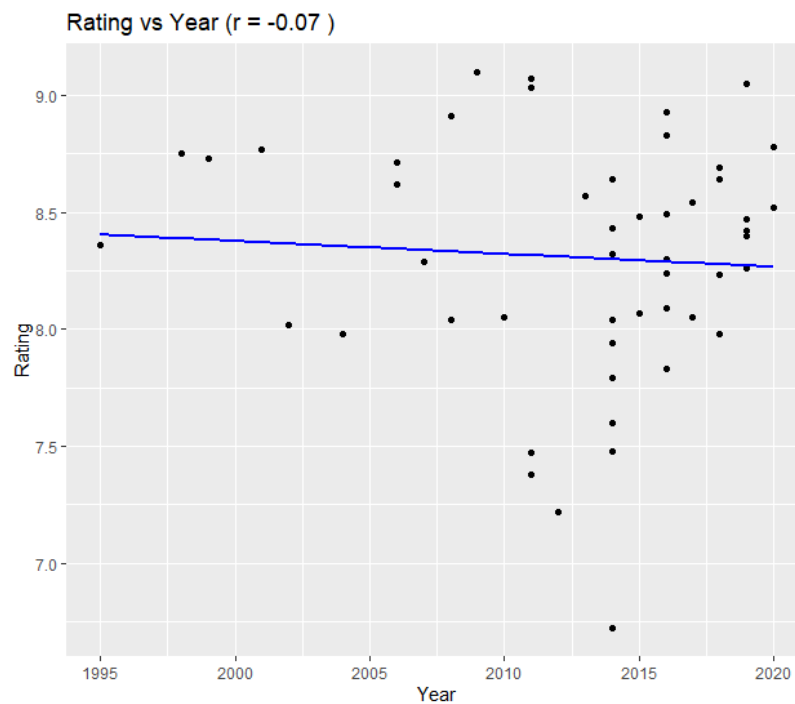
Exploratory Data Analysis (EDA) was conducted using visualizations to uncover patterns and relationships.



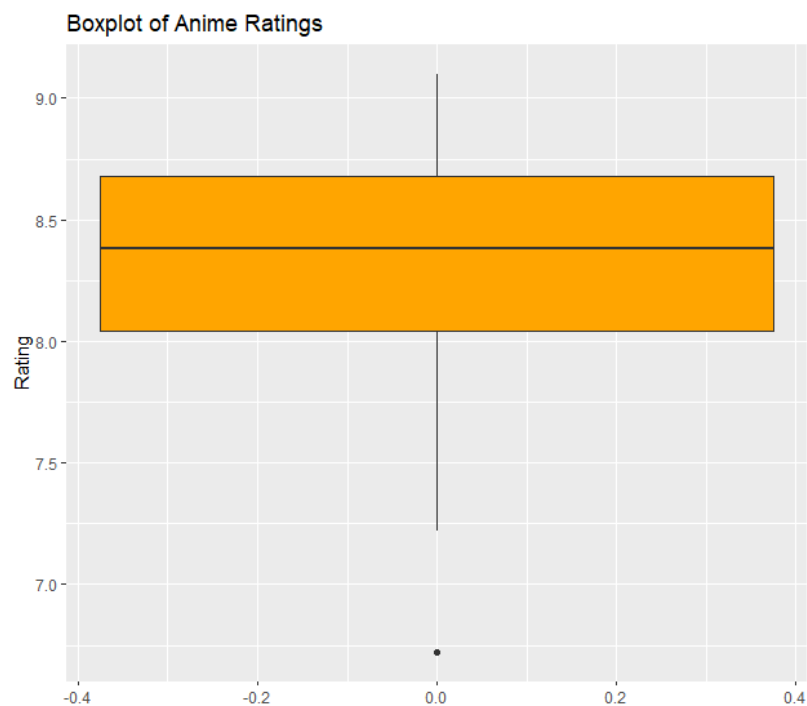
The Distribution of Anime Ratings shows that ratings are concentrated between 8.0 and 8.5, indicating that highly popular anime tend to receive consistently high scores, with very few rated below 8.0. This suggests a ceiling effect where top-ranked anime is generally well-received.



The Distribution of Anime Release Years reveals that the majority of top anime were produced after 2010, peaking around 2020. This reflects either a recency bias in popularity rankings or an increase in anime production and viewer engagement in recent years.



The Rating vs. Year scatter plot shows a near-zero correlation ($r = -0.07$), meaning release year alone does not strongly predict rating. Older anime are not necessarily rated higher or lower than newer ones in this sample.



The Boxplot of Anime Ratings confirms no extreme outliers, indicating that the rating distribution is relatively consistent without unusual skew.

Here's the whole coding process:

```
## Exploratory Data Analysis (EDA)

### Distribution of Ratings
ggplot(animas_df, aes(x = Rating)) +
  geom_histogram(binwidth = 0.1, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Anime Ratings", x = "Rating", y = "Count")

### Distribution of Years
ggplot(animas_df, aes(x = Year)) +
  geom_histogram(binwidth = 1, fill = "lightgreen", color = "black") +
  labs(title = "Distribution of Anime Release Years", x = "Year", y = "Count")

### Ratings vs. Year (Scatter Plot)
ggplot(animas_df, aes(x = Year, y = Rating)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(title = paste("Rating vs Year (r =",
                    round(cor(animas_df$Year, animas_df$Rating), 2), ")"),
        x = "Year", y = "Rating")

### Detecting Outliers (using Box Plot)
ggplot(animas_df, aes(y = Rating)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Boxplot of Anime Ratings", y = "Rating")
```

4. Data Preprocessing

Several preprocessing steps were applied to prepare the data for modelling.

```
# 4. Data Pre-Processing

## Checking if there's any Missing Data
colsums(is.na(animas_df))

## Checking how old the Anime is,
current_year <- as.integer(format(Sys.Date(), "%Y"))
animas_df$Anime_Age_in_Years <- (current_year - animas_df$Year)
animas_df

## Excluding the Title for Feature selection (For Numeric Analysis)
animas_numeric_df <- animas_df
animas_numeric_df$Titles <- NULL
animas_numeric_df

## Encoding Categorical Variable
animas_df$Titles <- as.factor(animas_df$Titles)
animas_df

## Applying Min-Max Normalization
animas_scaled <- as.data.frame(scale(animas_numeric_df))
animas_scaled

head(animas_scaled)
```

First, a new feature, **Anime_Age_in_Years**, was created by subtracting the release year from the current year (2026). This transforms the release year into a more interpretable measure of how long the anime has been available, which may influence its rating through factors like nostalgia or dated content. Next, categorical titles were converted into factors for potential use in categorical encoding, though they were later excluded from the numeric model to focus on continuous predictors. The dataset was then split into **training (70%)** and **testing (30%)** sets to evaluate model performance on unseen data. Finally, Min-Max normalization was applied to numeric features to ensure that variables like **Rank** and **Anime_Age_in_Years** were on a comparable scale, preventing any single feature from disproportionately influencing the model due to its magnitude.

```
> head(animas_scaled)
  Rank      Year      Rating Anime_Age_in_Years
1 -1.680686  0.06270063  0.5155085         -0.06270063
2 -1.612087 -1.09231103  0.6132908          1.09231103
3 -1.543487 -0.59730603  1.5520012          0.59730603
4 -1.474888  0.39270397  0.3395003         -0.39270397
5 -1.406288  1.05271063  0.2221615        -1.05271063
6 -1.337689 -0.10230103 -2.1246145          0.10230103
```

5. Modelling and Analysis

A linear regression model was selected to predict Rating based on Rank, Year, and Anime_Age_in_Years. Linear regression is appropriate here because the target variable is continuous and the relationships are assumed to be linear.

```
> ## Check correlation between variables
> cor_matrix <- cor(model_data)
> cor_matrix
```

	Rank	Year	Rating	Anime_Age_in_Years
Rank	1.00000000	-0.04839459	-0.17905753	0.04839459
Year	-0.04839459	1.00000000	-0.06538256	-1.00000000
Rating	-0.17905753	-0.06538256	1.00000000	0.06538256
Anime_Age_in_Years	0.04839459	-1.00000000	0.06538256	1.00000000

```
> summary(lm_model)
```

Call:
lm(formula = Rating ~ ., data = train_data)

Residuals:

Min	1Q	Median	3Q	Max
-1.4077	-0.2906	0.1366	0.3399	0.6374

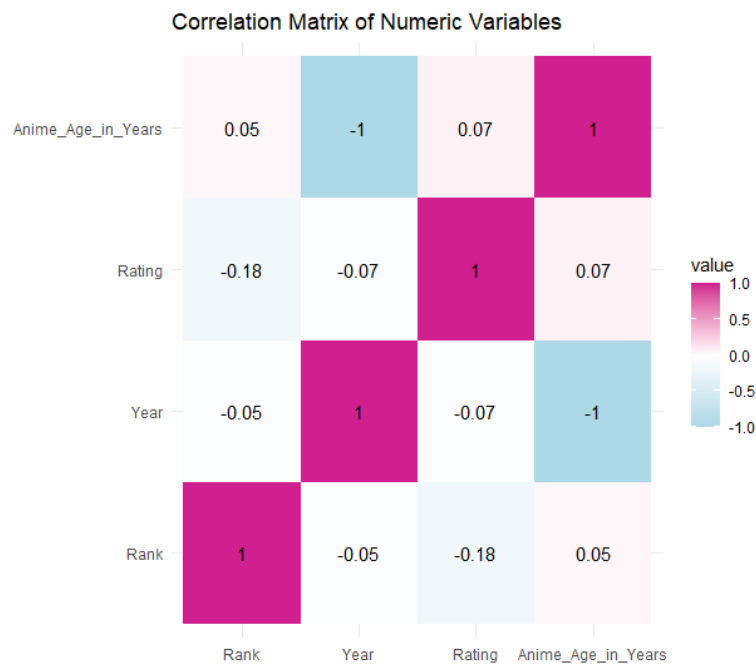
Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	22.023853	27.410843	0.803	0.4276
Rank	-0.010953	0.005943	-1.843	0.0746 .
Year	-0.006682	0.013617	-0.491	0.6270
Anime_Age_in_Years	NA	NA	NA	NA

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4956 on 32 degrees of freedom
Multiple R-squared: 0.1025, Adjusted R-squared: 0.04638
F-statistic: 1.827 on 2 and 32 DF, p-value: 0.1773

```
> ##### Mean Absolute Error (MAE)
> mae <- mean(abs(predictions - actual))
> mae
[1] 0.4272801
>
> ##### Root Mean Squared Error (RMSE)
> rmse <- sqrt(mean((predictions - actual)^2))
> rmse
[1] 0.5591582
>
> ##### R-squared (on test data)
> ss_res <- sum((actual - predictions)^2)
> ss_res
[1] 4.689868
> ss_tot <- sum((actual - mean(actual))^2)
> ss_tot
[1] 4.039773
> r_squared_test <- 1 - (ss_res / ss_tot)
> r_squared_test
[1] -0.1609236
```



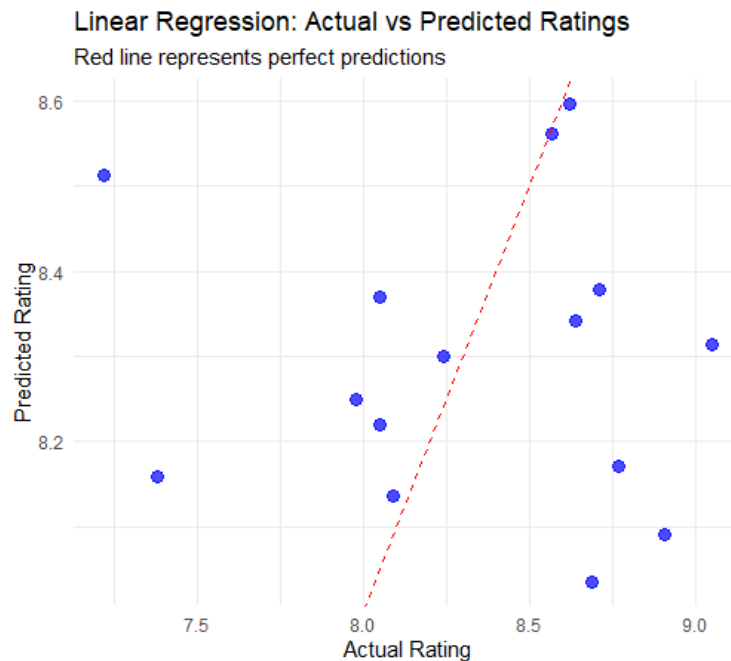
The correlation matrix heatmap revealed weak relationships between predictors and the target, with the strongest being a mild negative correlation between Rank and Rating (-0.18), meaning higher-ranked (more popular) anime tend to have slightly higher ratings. However, the correlations are generally low, suggesting limited predictive power.

```
> results_df
```

	Actual	Predicted	ID
1	8.57	8.561604	1
2	8.62	8.597426	2
6	7.22	8.513519	3
16	8.05	8.370574	4
18	8.64	8.341985	5
20	9.05	8.313396	6
22	8.71	8.378358	7
23	8.24	8.300582	8
34	8.05	8.220188	9
35	7.98	8.249328	10
38	8.09	8.136281	11
39	7.38	8.158739	12
44	8.77	8.170794	13
46	8.69	8.035289	14
47	8.91	8.091158	15

The model was trained on the training set and evaluated on the test set. Performance metrics were calculated: **Mean Absolute Error (MAE)** was **0.428**, meaning the model's predictions were off by about **0.43** points on average on the rating scale. **Root Mean Squared Error (RMSE)** was **0.559**, which penalizes larger errors more heavily and indicates moderate prediction inaccuracy. Most notably, **R-squared (R²)** was **-0.161**, which is below zero. A

negative R^2 means that the model performs worse than simply predicting the mean rating for all observations. This occurs when the model does not capture the underlying data structure and is essentially fitting noise. This visualization clearly illustrates the model's poor fit. The red line represents perfect predictions (where predicted equals actual), but the blue points (model predictions) are scattered widely and show no clear alignment with the line. Predictions do not trend upward with actual ratings, confirming the model's inability to capture the rating pattern.



Here's the whole coding process:

```
## visualize correlations
melted_cor <- melt(cor_matrix)
ggplot(melted_cor, aes(var1, var2, fill = value)) +
  geom_tile() +
  scale_fill_gradient2(low = "lightblue", high = "violetred", mid = "white", midpoint = 0) +
  geom_text(aes(label = round(value, 2)), size = 4) +
  labs(title = "Correlation Matrix of Numeric Variables", x = "", y = "") +
  theme_minimal()
```

```
### Linear Regression (Predict Rating)

#### Split data into training (70%) and testing (30%) sets
set.seed(123)
train_index <- sample(1:nrow(model_data), 0.7 * nrow(model_data))
train_data <- model_data[train_index, ]
test_data <- model_data[-train_index, ]

#### Predicting Rating based on Rank, Year, and Anime_Age_in_Years
lm_model <- lm(Rating ~ ., data = train_data)
lm_model
summary(lm_model)
```

```

#### Make predictions on test set
predictions <- predict(lm_model, newdata = test_data)

#### Calculate performance metrics
actual <- test_data$Rating
actual

#### Mean Absolute Error (MAE)
mae <- mean(abs(predictions - actual))
mae

#### Root Mean Squared Error (RMSE)
rmse <- sqrt(mean((predictions - actual)^2))
rmse

#### R-squared (on test data)
ss_res <- sum((actual - predictions)^2)
ss_res
ss_tot <- sum((actual - mean(actual))^2)
ss_tot
r_squared_test <- 1 - (ss_res / ss_tot)
r_squared_test

#### Visualize predictions vs actual
results_df <- data.frame(
  Actual = actual,
  Predicted = predictions,
  ID = 1:length(actual)
)
results_df

```

```

ggplot(results_df, aes(x = Actual, y = Predicted)) +
  geom_point(color = "blue", size = 3, alpha = 0.7) +
  geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
  labs(
    title = "Linear Regression: Actual vs Predicted Ratings",
    subtitle = "Red line represents perfect predictions",
    x = "Actual Rating",
    y = "Predicted Rating"
  ) +
  theme_minimal()

```

6. Conclusion and Future Work

This study explored the relationship between popularity rank, release year, and user ratings for the top 50 anime on MyAnimeList. While EVA revealed interesting trends—such as the concentration of high ratings and the prevalence of recent anime in the popularity rankings—the predictive modelling effort was unsuccessful. The linear regression model not only failed to explain variance in ratings but performed worse than a simple mean predictor, as evidenced by the negative R^2 . The findings suggest that anime ratings are complex and multifactorial, influenced by elements beyond rank and release year, and that a simple linear approach is inadequate for this prediction task.

Key limitations include the small sample size (only 50 entries), the limited feature set, and the static nature of the data (single scraping instance). Additionally, the model assumes linear relationships, which may not hold in practice. The dataset also excludes important qualitative factors such as genre, director, studio, and user review sentiment, which are likely critical drivers of rating variation.

Future research could expand the dataset to include hundreds or thousands of anime, incorporate additional features like genre, episode count, and production studio, and employ more advanced modelling techniques such as Random Forests or Gradient Boosting Machines, which can capture nonlinear relationships and interactions. Text mining of user reviews could also be integrated to perform sentiment analysis, providing a richer understanding of what drives high ratings. Longitudinal analysis tracking rating changes over time could further enhance predictive accuracy and insight into anime reception dynamics. Additionally, exploring clustering techniques to group anime by characteristics before modelling may yield more interpretable and accurate results.